

2344.TW - Winbond Electronics | Full Deep Dive

September 21, 2026 | Framework 6.0.2 | NOT HELD | BOTTLENECK - specialty DRAM and code-storage supply | High Conviction

Analytical Base: 48/60 (80%). Valuation: 6x forward four-quarter owner-attributable operating-profit proxy after Nuvoton minority interests.

Snapshot: TWD 174.00 per share on September 21, 2026 close; 4,720.000 million shares, implying TWD 821,280.00 million equity value. Source [W7]. Price is a dated header snapshot, not a valuation input in the body.

Active horizon Q3 2027 ceiling: TWD 447.15/share. Q1 2028: TWD 471.88/share.

All financial model amounts are TWD millions, shares in millions, unless stated. Calendar and fiscal quarters coincide. Actual financials stop at Q2; July/August monthly revenue is Preliminary. Q3 2026 onward is Model. OP in modeled valuation tables means owner-attributable OP proxy, not reported consolidated IFRS OP.

1. Thesis, bottleneck and expanded TAM

Winbond manufactures customized/specialty DRAM and code-storage flash, with a consolidated logic business through Nuvoton. It enables reliable boot/code storage and working memory in computers, communications, industrial equipment, cars and edge devices. Framework DB6 provides the current supply-cycle driver; physical AI widens the application set.

HBM and leading-edge DRAM investment can crowd out mature products, while exits from low-density 2D NAND support SLC supply scarcity. NOR and SLC NAND are distinct from high-layer TLC/QLC used in enterprise SSDs. Higher eSSD demand is an indirect allocation tailwind and can raise DRAM/controller content; it does not make all Winbond flash an enterprise-NAND product.

The established business is in a price/qualification bottleneck phase. Low power, long product life, automotive qualification and an owned fab help retention. CUBE custom memory and silicon capacitors represent earlier design-in opportunities for edge architectures. They are not proof of a mass-market accelerator-HBM position. Physical-IP advantages merit a strong thesis, but competing qualified suppliers prevent a permanent sole-source classification. [W2,W3,W8]

The 2025 annual report competitive-niche sections were reviewed visually. They identify expanding Chinese supply and pricing pressure as real threats, alongside Winbond process/customization and customer-longevity advantages. The Q2 presentation identifies a 2025 NOR market of USD3.4B with Winbond at 23%, versus GigaDevice/Macronix 17% each and Infineon/Micron among other suppliers. Its internally estimated SLC NAND market was USD1.5B with Winbond at 15%. These are historical scope anchors, not 2026 guidance. Nuvoton also faces MCU/MPU competition. No new undercovered listed monopoly was verified in this limited peer screen. [W2,W5]

TAM expansion and S-curve

Phase	Economic exposure	Model treatment
Phase 1: current infrastructure and embedded electronics	Specialty DRAM, NOR and SLC code/data storage	100% established demand; separate CMS and flash gross-profit pools.
Phase 2: physical AI, 2026-30	Automotive, industrial, robotic and edge-inference memory	50% conceptual confidence; higher density/qualification supports the existing volume and mix path.
Phase 3: new architectures, 2027+	CUBE custom memory and Si-Cap integration	20% conceptual confidence for expanded applications. No unsupported fixed dollar TAM; timed contribution is embedded in CMS allocation.

Use a calibrated Plan-B product-volume and price model because named customer capex/wallet shares are not disclosed. Q2 segment anchors from H1 less Q1 are CMS 31,744.098; Flash 19,955.706; Logic 7,734.608; Other/software 408.853, totaling 59,843.265. Monthly July/August revenue then constrains Q3 rather than stale consensus. Demand capacity is tied to current

utilization, Module A/16nm and disclosed NAND bit-growth expectations. A USD2T all-memory forecast is not Winbond available TAM; its specialty pools are much narrower. [W1,W1a,W2,W4]

2. Latest evidence and financial anchor

May 5 and August 6 management presentations were read, including product and outlook sections. Full management Q&A transcripts could not be retrieved from the webcast pages. This remains an earnings-protocol gap; no AI-generated call summary substitutes for primary management language. [W2,W3]

July revenue 26,773.179 plus August 27,308.717 and modeled September at August x 1.02 (27,854.891) gives Q3 revenue 81,936.787. This is a monthly-constrained forecast, not issued Q3 guidance. The CMS ASP is the residual required after separately modeled flash, logic and software; it must be replaced when segment actuals arrive. [W4]

Management sees further Q3 contract-price room, growing LTA coverage, meaningful 16nm contributions beginning in 2027 and CUBE/Si-Cap growth after 2027. NAND bit-growth targets exceed 20% in 2026 and 40% in 2027. Module A output expansion precedes Module B: reported management timing puts B tool-in around early 2029 and production near end-2029, not 2027. [W2,W6]

Period	Status	Revenue	Gross profit	Operating profit	Source
Q1 2026	Actual	38,253.06	20,415.00	12,550.49	W1a; consolidated IFRS OP
Q2 2026	Actual	59,843.26	39,645.28	28,979.34	W1; consolidated IFRS OP

Q2 gross profit 39,645.280 less R&D 6,509.088, SG&A 3,830.352 and expected-credit loss 326.504 = consolidated OP 28,979.336. Memory GP is 36,337 (rounded presentation). Use a transparent 75% CMS GM assumption; Flash GM solves to 62.7837% so CMS plus Flash GP matches that anchor. Logic GM uses the reported rounded 38%; residual/software GM solves to 90.2840% to preserve consolidated GP exactly. These allocations are proxies where detailed GP is unavailable. Segment profit in the filing already includes allocated costs and is not mislabeled as gross profit. [W1,W1a,W2]

No numeric Q3 consolidated revenue or OP guide was located. Q2 CMS bit shipments declined in the low teens while USD ASP roughly doubled; Flash bits rose in the mid-teens and ASP in the low-40% range. Those extraordinary changes are not repeated mechanically. Q3 is anchored to actual monthly sales, with future increases decelerating. [W2,W4]

Balance sheet and capital allocation

June consolidated cash was 43,014.862 plus 18,811.415 classified as hedging financial assets, or 61,826.277 in the cash-flow presentation. Debt included 3,023.703 short-term, 43,085.137 current long-term/bonds and about 22,531.642 non-current borrowings/bonds, before leases. Thus the presentation cash number cannot be compared only with non-current bank debt. The subsequent parent ECB conversion changes both debt and shares. Q2 CFO was 23,306; cash capex 5,172. Updated memory capex plan is 39,500 for 2026, versus H1 memory spending 7,500. Module B adds later commitments. No net-cash amount is added to the ceiling. [W1,W2]

3. Base score and evidence ledger

Category	Score	Evidence
Core Thesis /20	17	Qualification, fab/process control, longevity and code-storage leadership; several capable competitors.
Cycle Stage /10	9	Pricing and utilization confirmed; 16nm and NAND capacity extend the near-term cycle.
TAM Expansion /10	9	Memory and secure code storage span infrastructure, vehicles and physical AI; custom memory adds a credible path.
Catalysts /5	5	Monthly revenue, 16nm 2027, higher-density SLC ramp and design-in progression.
Macro /10	9	AI allocation and mature-product supply exits support scarcity; end demand still matters.
Alpha /5	3	Local segment/ownership and bit-growth evidence; broader memory thesis widely followed.
Risks	-4	Competition, execution, product/customer-cycle concentration and capital/dilution.
Total	48/60 = 80%	High Conviction

48/60 = 80% is an independently derived Base for this new entry. Growing LTA coverage improves visibility; no confirmed customer cash prepayment is assumed and no prepayment bonus is awarded. Geographic location alone is neutral.

4. Quarterly operating model

Four revenue/GP vectors: CMS, Flash, Logic and Other/software. NOR and SLC prices/volumes are calculated separately inside the Flash bridge because no separate reported gross-margin pools exist. Every expense ratio is visible. Consolidated OP is then reduced by the Nuvoton minority operating-profit proxy before applying 6x to Winbond diluted shares. The ownership adjustment is not an IFRS operating expense.

Revenue vector	Driver and source	Confidence
CMS / specialty DRAM	W1-W4: CMS actual / monthly anchor; analyst volume, ASP and cost	MEDIUM monthly revenue / LOW mix, margin and longer term
Flash: NOR and SLC NAND	W1,W2,W3: Flash actual; NOR/SLC split proxy; U1 NAND indices	MEDIUM monthly revenue / LOW mix, margin and longer term
Logic / Nuvoton	W1,W2: consolidated logic sales; 38% rounded GM	MEDIUM monthly revenue / LOW mix, margin and longer term
Other / software	W1/W1a reported residual; calibrated proxy gross margin	MEDIUM monthly revenue / LOW mix, margin and longer term

For each vector revenue = Q2 revenue x effective volume index x realized-ASP index; GM = 1 - (1 - anchor GM) x unit-cost index / ASP index. Q3 CMS volume is +4%; the calibrated CMS ASP index is 1.476270. Further CMS ASP increases are 12% Q4 and 10%/5%/3%/0% through 2027, analyst estimates. Flash uses a 75% NOR / 25% SLC NAND base revenue mix proxy: 2025 market-share math implies roughly 22% NAND before the 24nm ramp, but no exact 2026 split is disclosed. NOR ASP starts +20% Q3, then +10% Q4 and +8%/+4%/+2%/0% in 2027. Only SLC uses the supplied NAND +23%/+20%/+15% schedule; Q2/Q3/Q4 2027 then +5%/+2%/0%. The broad NAND index is an imperfect SLC proxy, not verified company contract guidance. Weighted Flash revenue = anchor x (75% x NOR volume x NOR ASP + 25% x NAND volume x NAND ASP); its displayed ASP divides this sum by weighted volume. This prevents a blended-index cross-term error. CMS and NOR ASP decrease 5% cumulatively in 2028, NAND 13%; 2029/30 recover 3%/2%. CMS unit cost rises for the 2027 ramp and again with B in late 2029. Flash costs improve after the 24nm ramp. Logic margins rise gradually from 38.5% to 42.5%; software margin stays at its calibrated proxy. Ratios and product-mix paths are analyst assumptions, with neutral FX.

Physical and commercial drivers

Quarter	CMS volume	CMS ASP	NOR volume / ASP	NAND volume / ASP	Flash effective volume / ASP
Q3 2026	1.0400	1.47627	1.020 / 1.2000	1.060 / 1.2300	1.0300 / 1.20772
Q4 2026	1.0800	1.65342	1.040 / 1.3200	1.120 / 1.4760	1.0600 / 1.36121
Q1 2027	1.1300	1.81877	1.080 / 1.4256	1.240 / 1.6974	1.1200 / 1.50083
Q2 2027	1.1900	1.90970	1.120 / 1.4826	1.360 / 1.7823	1.1800 / 1.56896
Q3 2027	1.2500	1.96699	1.160 / 1.5123	1.480 / 1.8179	1.2400 / 1.60348
Q4 2027	1.3100	1.96699	1.200 / 1.5123	1.600 / 1.8179	1.3000 / 1.60632
Q1 2028	1.3700	1.94193	1.240 / 1.4930	1.690 / 1.7557	1.3525 / 1.57507
Q2 2028	1.4300	1.91719	1.280 / 1.4740	1.780 / 1.6956	1.4050 / 1.54419
Q3 2028	1.4900	1.89276	1.320 / 1.4552	1.870 / 1.6376	1.4575 / 1.51371
Q4 2028	1.5500	1.86864	1.360 / 1.4367	1.960 / 1.5816	1.5100 / 1.48369
Q1 2029	1.6100	1.88250	1.400 / 1.4473	2.050 / 1.5933	1.5625 / 1.49521
Q2 2029	1.6700	1.89647	1.440 / 1.4581	2.140 / 1.6051	1.6150 / 1.50678
Q3 2029	1.7300	1.91053	1.480 / 1.4689	2.230 / 1.6170	1.6675 / 1.51841
Q4 2029	1.8300	1.92470	1.530 / 1.4798	2.350 / 1.6290	1.7350 / 1.53031
Q1 2030	1.9300	1.93426	1.590 / 1.4871	2.490 / 1.6371	1.8150 / 1.53856
Q2 2030	2.0300	1.94386	1.650 / 1.4945	2.630 / 1.6452	1.8950 / 1.54679
Q3 2030	2.1300	1.95350	1.710 / 1.5019	2.770 / 1.6534	1.9750 / 1.55503
Q4 2030	2.2300	1.96320	1.770 / 1.5094	2.910 / 1.6616	2.0550 / 1.56326

Qualification and product allocation

CMS volume contains qualified capacity and weighted custom/beyond-current-fab increments. These are allocations within one physical output path, not additional independent revenue. Customer design-in progression must precede recognized sales.

Source pool	Unweighted increment	Recognition weight / timing	Treatment
CUBE / Si-Cap custom mix	CMS index 0.05 / 0.0833 / 0.1167 / 0.15 in 2027	60% management-indicated ramp	0.03 / 0.05 / 0.07 / 0.09 included within CMS 1.13 / 1.19 / 1.25 / 1.31; no second premium.
Module B capacity	Additional CMS indices 0.10 Q4 2029; 0.175 / 0.25 / 0.325 / 0.40 in 2030	80%, only from Q4 2029	Included increments 0.08 then 0.14 / 0.20 / 0.26 / 0.32. Later dates are LOW confidence.
NAND product migration	Company >40% 2027 bit-growth direction	Established product path, model allocation	Average 2027 NAND volume index 1.42; mix within Flash, no separate all-memory TAM add-on.

Vector revenue, gross margin and gross profit

For each row: vector gross profit = vector revenue x its own gross margin. The following tables expose the inputs; they are analyst model assumptions unless explicitly identified as reported or guided.

CMS / specialty DRAM

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	48,737.39	82.39%	40,153.78
Q4 2026	56,685.33	83.67%	47,428.75
Q1 2027	65,240.62	84.47%	55,107.11
Q2 2027	72,139.96	84.95%	61,279.51
Q3 2027	78,050.58	85.26%	66,543.35
Q4 2027	81,797.01	85.13%	69,633.47
Q1 2028	84,453.49	84.81%	71,624.11
Q2 2028	87,029.00	84.74%	73,751.23
Q3 2028	89,525.16	84.68%	75,808.54
Q4 2028	91,943.59	84.61%	77,797.62
Q1 2029	96,211.04	84.73%	81,517.49
Q2 2029	100,536.75	84.84%	85,295.61
Q3 2029	104,921.32	84.95%	89,132.60
Q4 2029	111,809.33	84.41%	94,381.82
Q1 2030	118,504.36	84.23%	99,818.19
Q2 2030	125,263.08	84.57%	105,930.93
Q3 2030	132,085.97	84.90%	112,139.57
Q4 2030	138,973.50	85.23%	118,444.59

Flash: NOR and SLC NAND

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	24,823.90	68.57%	17,021.33
Q4 2026	28,793.69	71.57%	20,606.41
Q1 2027	33,544.14	73.72%	24,727.07
Q2 2027	36,945.51	74.86%	27,656.10
Q3 2027	39,678.11	75.63%	30,008.45
Q4 2027	41,671.80	75.90%	31,630.80
Q1 2028	42,511.36	75.43%	32,064.85
Q2 2028	43,295.59	75.18%	32,547.93
Q3 2028	44,027.06	74.92%	32,986.04
Q4 2028	44,708.23	74.67%	33,381.65
Q1 2029	46,621.71	75.11%	35,017.36
Q2 2029	48,561.11	75.30%	36,566.86
Q3 2029	50,526.72	75.49%	38,142.56
Q4 2029	52,984.10	74.71%	39,583.21
Q1 2030	55,725.93	74.36%	41,437.55
Q2 2030	58,493.67	74.74%	43,716.24
Q3 2030	61,287.51	75.11%	46,032.91
Q4 2030	64,107.65	75.48%	48,387.76

Logic / Nuvoton

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	7,966.65	38.50%	3,067.16
Q4 2026	8,353.38	39.00%	3,257.82
Q1 2027	8,353.38	39.50%	3,299.58
Q2 2027	8,740.11	40.00%	3,496.04
Q3 2027	9,126.84	40.50%	3,696.37
Q4 2027	9,513.57	41.00%	3,900.56
Q1 2028	9,822.95	41.50%	4,076.53
Q2 2028	10,132.34	41.50%	4,204.92
Q3 2028	10,441.72	41.50%	4,333.31
Q4 2028	10,751.11	41.50%	4,461.71
Q1 2029	11,060.49	42.00%	4,645.41
Q2 2029	11,369.87	42.00%	4,775.35
Q3 2029	11,679.26	42.00%	4,905.29
Q4 2029	11,988.64	42.00%	5,035.23
Q1 2030	12,298.03	42.50%	5,226.66
Q2 2030	12,607.41	42.50%	5,358.15
Q3 2030	12,916.80	42.50%	5,489.64
Q4 2030	13,226.18	42.50%	5,621.13

Other / software

Quarter	Revenue	Gross margin	Gross profit	
Q3 2026		408.85	90.28%	369.13
Q4 2026		421.12	90.28%	380.20
Q1 2027		425.21	90.28%	383.89
Q2 2027		437.47	90.28%	394.97
Q3 2027		449.74	90.28%	406.04
Q4 2027		462.00	90.28%	417.12
Q1 2028		474.27	90.28%	428.19
Q2 2028		486.54	90.28%	439.26
Q3 2028		498.80	90.28%	450.34
Q4 2028		511.07	90.28%	461.41
Q1 2029		523.33	90.28%	472.49
Q2 2029		535.60	90.28%	483.56
Q3 2029		547.86	90.28%	494.63
Q4 2029		560.13	90.28%	505.71
Q1 2030		572.39	90.28%	516.78
Q2 2030		584.66	90.28%	527.85
Q3 2030		596.93	90.28%	538.93
Q4 2030		609.19	90.28%	550.00

Combined quarterly earnings

Quarter	Status	Revenue	Gross profit	R&D	SG&A	Other / NCI	Owner OP	OP margin
Q3 2026	Model	81,936.79	60,611.40	7,783.99	4,916.21	165.13	47,746.06	58.27%
Q4 2026	Model	94,253.52	71,673.18	8,482.82	5,466.70	249.13	57,474.53	60.98%
Q1 2027	Model	107,563.35	83,517.66	9,142.88	6,023.55	345.00	68,006.23	63.22%
Q2 2027	Model	118,263.05	92,826.62	9,697.57	6,504.47	428.45	76,196.13	64.43%
Q3 2027	Model	127,305.27	100,654.21	10,184.42	6,874.48	509.22	83,086.08	65.27%
Q4 2027	Model	133,444.39	105,581.94	10,408.66	7,072.55	582.33	87,518.40	65.58%
Q1 2028	Model	137,262.07	108,193.68	10,843.70	7,137.63	605.73	89,606.62	65.28%
Q2 2028	Model	140,943.46	110,943.35	11,134.53	7,329.06	622.57	91,857.19	65.17%
Q3 2028	Model	144,492.75	113,578.23	11,414.93	7,513.62	638.93	94,010.75	65.06%
Q4 2028	Model	147,913.99	116,102.39	11,685.21	7,691.53	654.86	96,070.80	64.95%
Q1 2029	Model	154,416.57	121,652.75	12,044.49	7,875.25	689.02	101,043.99	65.44%
Q2 2029	Model	161,003.33	127,121.38	12,558.26	8,211.17	715.09	105,636.86	65.61%
Q3 2029	Model	167,675.16	132,675.09	13,078.66	8,551.43	741.42	110,303.57	65.78%
Q4 2029	Model	177,342.20	139,505.97	13,832.69	9,044.45	776.74	115,852.09	65.33%
Q1 2030	Model	187,100.71	146,999.18	14,406.75	9,355.04	906.46	122,330.93	65.38%
Q2 2030	Model	196,948.82	155,533.17	15,165.06	9,847.44	944.69	129,575.97	65.79%
Q3 2030	Model	206,887.21	164,201.05	15,930.32	10,344.36	983.19	136,943.18	66.19%
Q4 2030	Model	216,916.52	173,003.48	16,702.57	10,845.83	1,021.96	144,433.12	66.58%

Operating-profit ownership reconciliation

Other/NCI combines credit cost plus minority OP claim. Consolidated OP = GP minus R&D, SG&A and credit cost. Owner OP = consolidated OP minus minority claim. A negative claim adds back minority losses. These are proxy calculations, not IFRS attribution.

Quarter	Credit cost	Consolidated OP	Logic OP proxy	Minority OP claim	Owner OP
Q3 2026	327.75	47,583.45	-318.67	-162.62	47,746.06
Q4 2026	377.01	57,346.64	-250.60	-127.88	57,474.53
Q1 2027	430.25	67,920.97	-167.07	-85.25	68,006.23
Q2 2027	473.05	76,151.53	-87.40	-44.60	76,196.13

Quarter	Credit cost	Consolidated OP	Logic OP proxy	Minority OP claim	Owner OP
Q3 2027	509.22	83,086.08	0.00	0.00	83,086.08
Q4 2027	533.78	87,566.95	95.14	48.55	87,518.40
Q1 2028	480.42	89,731.93	245.57	125.32	89,606.62
Q2 2028	493.30	91,986.45	253.31	129.26	91,857.19
Q3 2028	505.72	94,143.96	261.04	133.21	94,010.75
Q4 2028	517.70	96,207.96	268.78	137.16	96,070.80
Q1 2029	463.25	101,269.76	442.42	225.77	101,043.99
Q2 2029	483.01	105,868.94	454.79	232.08	105,636.86
Q3 2029	503.03	110,541.96	467.17	238.40	110,303.57
Q4 2029	532.03	116,096.80	479.55	244.71	115,852.09
Q1 2030	561.30	122,676.09	676.39	345.16	122,330.93
Q2 2030	590.85	129,929.82	693.41	353.85	129,575.97
Q3 2030	620.66	137,305.71	710.42	362.53	136,943.18
Q4 2030	650.75	144,804.33	727.44	371.21	144,433.12

Quarterly drivers and confidence

Calendar / fiscal	Status	Revenue	OP margin	OP	Main driver	Confidence
Q3 2026 / FQ3 FY2026	Model	81,936.79	58.27%	47,746.06	Two monthly actuals; mix proxy	MEDIUM revenue / LOW margin
Q4 2026 / FQ4 FY2026	Model	94,253.52	60.98%	57,474.53	Code-storage and DRAM repricing	MEDIUM revenue / LOW margin
Q1 2027 / FQ1 FY2027	Model	107,563.35	63.22%	68,006.23	16nm and weighted custom ramp	MEDIUM revenue / LOW margin
Q2 2027 / FQ2 FY2027	Model	118,263.05	64.43%	76,196.13	NAND density mix	MEDIUM revenue / LOW margin
Q3 2027 / FQ3 FY2027	Model	127,305.27	65.27%	83,086.08	Qualified bits and pricing	MEDIUM revenue / LOW margin
Q4 2027 / FQ4 FY2027	Model	133,444.39	65.58%	87,518.40	Cost absorption	MEDIUM revenue / LOW margin
Q1 2028 / FQ1 FY2028	Model	137,262.07	65.28%	89,606.62	Price normalization; installed output	LOW
Q2 2028 / FQ2 FY2028	Model	140,943.46	65.17%	91,857.19	Price normalization; installed output	LOW
Q3 2028 / FQ3 FY2028	Model	144,492.75	65.06%	94,010.75	Price normalization; installed output	LOW
Q4 2028 / FQ4 FY2028	Model	147,913.99	64.95%	96,070.80	Price normalization; installed output	LOW
Q1 2029 / FQ1 FY2029	Model	154,416.57	65.44%	101,043.99	Existing output growth	LOW
Q2 2029 / FQ2 FY2029	Model	161,003.33	65.61%	105,636.86	Existing output growth	LOW
Q3 2029 / FQ3 FY2029	Model	167,675.16	65.78%	110,303.57	Existing output growth	LOW
Q4 2029 / FQ4 FY2029	Model	177,342.20	65.33%	115,852.09	Weighted Module B first output	LOW
Q1 2030 / FQ1 FY2030	Model	187,100.71	65.38%	122,330.93	Module B conditional ramp	LOW
Q2 2030 / FQ2 FY2030	Model	196,948.82	65.79%	129,575.97	Module B conditional ramp	LOW
Q3 2030 / FQ3 FY2030	Model	206,887.21	66.19%	136,943.18	Module B conditional ramp	LOW
Q4 2030 / FQ4 FY2030	Model	216,916.52	66.58%	144,433.12	Module B conditional ramp	LOW

Annual trajectory cross-check

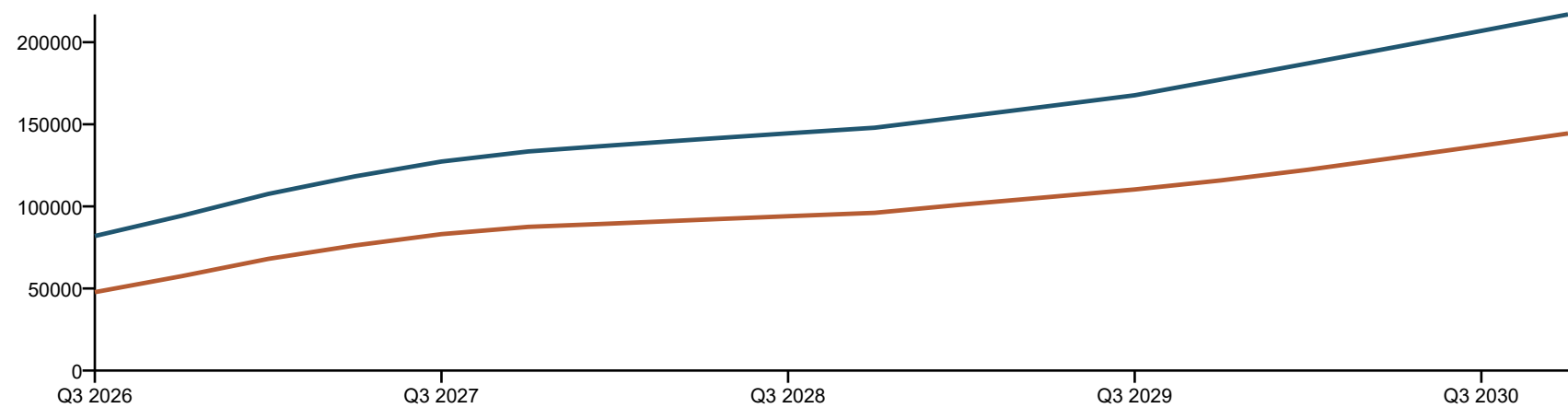
The calendar-bucket sums below are a capacity and demand cross-check; quarterly windows remain the live valuation method. Years beyond the latest guidance are model extensions, not company guidance.

Year	Revenue	Operating profit	OP margin
2027	486,576.06	314,806.84	64.70%
2028	570,612.27	371,545.35	65.11%
2029	660,437.26	432,836.51	65.54%
2030	807,853.26	533,283.20	66.01%

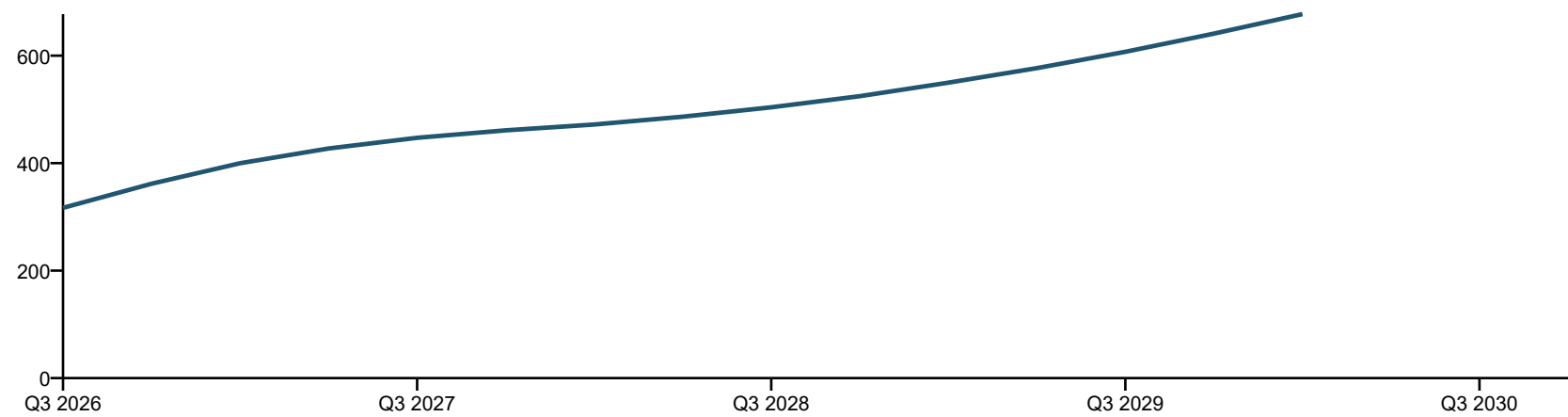
The 2027 average NAND bit index 1.42 versus a Q2 2026 anchor of 1 approximates the company direction, but is not a precise year-on-year guide comparison because the full 2026 bit denominator is undisclosed. Module B contributes only late in the horizon. Rising CMS/Flash share and price-cost spread lift consolidated margins while low-margin logic dilutes the mix. Modeled margins are conditional and unusually high; a commodity reversal would sharply lower them. Gross profit and OP are not free cash flow: the 39,500 capex plan and later construction consume cash. [W2]

5. Forward four-quarter ceiling and derivations

Revenue (blue), operating profit (orange) - TWD M



Forward-four-quarter ceiling - TWD/share



6x forward-four-quarter owner-attributable OP proxy is the user-selected multiple for Winbond as of September 21, 2026, held constant across all windows. The ownership bridge excludes profit economically belonging to Nuvoton minority holders. This proxy excludes interest, tax and nonoperating holdings; it is not attributable net income. Each one-turn change moves ceilings by 16.67%.

The framework convention applies the operating-profit multiple directly to diluted equity shares. It is neither a net-income P/E nor an enterprise-value bridge. No separate net-cash addition, tax conversion, probability haircut or price-dependent ranking enters this ceiling. Any probability-weighted revenue is already included once in its operating vector.

Ceiling quarter	Exact earnings window	Forward-4Q OP	Multiple	Diluted shares	Ceiling/share
Q3 2026	Q3 2026 to Q2 2027	249,422.94	6x	4,724.215	316.78
Q4 2026	Q4 2026 to Q3 2027	284,762.96	6x	4,724.215	361.66
Q1 2027	Q1 2027 to Q4 2027	314,806.84	6x	4,724.215	399.82
Q2 2027	Q2 2027 to Q1 2028	336,407.23	6x	4,724.215	427.25
Q3 2027	Q3 2027 to Q2 2028	352,068.29	6x	4,724.215	447.15
Q4 2027	Q4 2027 to Q3 2028	362,992.95	6x	4,724.215	461.02
Q1 2028	Q1 2028 to Q4 2028	371,545.35	6x	4,724.215	471.88
Q2 2028	Q2 2028 to Q1 2029	382,982.73	6x	4,724.215	486.41
Q3 2028	Q3 2028 to Q2 2029	396,762.40	6x	4,724.215	503.91
Q4 2028	Q4 2028 to Q3 2029	413,055.22	6x	4,724.215	524.60
Q1 2029	Q1 2029 to Q4 2029	432,836.51	6x	4,724.215	549.72
Q2 2029	Q2 2029 to Q1 2030	454,123.45	6x	4,724.215	576.76
Q3 2029	Q3 2029 to Q2 2030	478,062.56	6x	4,724.215	607.16
Q4 2029	Q4 2029 to Q3 2030	504,702.18	6x	4,724.215	641.00
Q1 2030	Q1 2030 to Q4 2030	533,283.20	6x	4,724.215	677.30

Recognition timing by ceiling

Ceiling quarter	Key inflection entering the forward window
Q3 2026	Two monthly actuals; mix proxy through NAND density mix
Q4 2026	Code-storage and DRAM repricing through Qualified bits and pricing
Q1 2027	16nm and weighted custom ramp through Cost absorption
Q2 2027	NAND density mix through Price normalization; installed output
Q3 2027	Qualified bits and pricing through Price normalization; installed output
Q4 2027	Cost absorption through Price normalization; installed output
Q1 2028	Price normalization; installed output through Price normalization; installed output
Q2 2028	Price normalization; installed output through Existing output growth
Q3 2028	Price normalization; installed output through Existing output growth
Q4 2028	Price normalization; installed output through Existing output growth
Q1 2029	Existing output growth through Weighted Module B first output
Q2 2029	Existing output growth through Module B conditional ramp
Q3 2029	Existing output growth through Module B conditional ramp
Q4 2029	Weighted Module B first output through Module B conditional ramp
Q1 2030	Module B conditional ramp through Module B conditional ramp

Complete derivation for every ceiling quarter

Q3 2026: OP window Q3 2026 to Q2 2027: $(47,746.06 + 57,474.53 + 68,006.23 + 76,196.13) \times 6 / 4,724.215 = \text{TWD } 316.78$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

Q4 2026: OP window Q4 2026 to Q3 2027: $(57,474.53 + 68,006.23 + 76,196.13 + 83,086.08) \times 6 / 4,724.215 = \text{TWD } 361.66$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

Q1 2027: OP window Q1 2027 to Q4 2027: $(68,006.23 + 76,196.13 + 83,086.08 + 87,518.40) \times 6 / 4,724.215 = \text{TWD } 399.82$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

Q2 2027: OP window Q2 2027 to Q1 2028: $(76,196.13 + 83,086.08 + 87,518.40 + 89,606.62) \times 6 / 4,724.215 = \text{TWD } 427.25$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

Q3 2027: OP window Q3 2027 to Q2 2028: $(83,086.08 + 87,518.40 + 89,606.62 + 91,857.19) \times 6 / 4,724.215 = \text{TWD } 447.15$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

Q4 2027: OP window Q4 2027 to Q3 2028: $(87,518.40 + 89,606.62 + 91,857.19 + 94,010.75) \times 6 / 4,724.215 = \text{TWD } 461.02$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

Q1 2028: OP window Q1 2028 to Q4 2028: $(89,606.62 + 91,857.19 + 94,010.75 + 96,070.80) \times 6 / 4,724.215 = \text{TWD } 471.88$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

Q2 2028: OP window Q2 2028 to Q1 2029: $(91,857.19 + 94,010.75 + 96,070.80 + 101,043.99) \times 6 / 4,724.215 = \text{TWD } 486.41$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

Q3 2028: OP window Q3 2028 to Q2 2029: $(94,010.75 + 96,070.80 + 101,043.99 + 105,636.86) \times 6 / 4,724.215 = \text{TWD } 503.91$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

Q4 2028: OP window Q4 2028 to Q3 2029: $(96,070.80 + 101,043.99 + 105,636.86 + 110,303.57) \times 6 / 4,724.215 = \text{TWD } 524.60$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

Q1 2029: OP window Q1 2029 to Q4 2029: $(101,043.99 + 105,636.86 + 110,303.57 + 115,852.09) \times 6 / 4,724.215 = \text{TWD } 549.72$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

Q2 2029: OP window Q2 2029 to Q1 2030: $(105,636.86 + 110,303.57 + 115,852.09 + 122,330.93) \times 6 / 4,724.215 = \text{TWD } 576.76$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

Q3 2029: OP window Q3 2029 to Q2 2030: $(110,303.57 + 115,852.09 + 122,330.93 + 129,575.97) \times 6 / 4,724.215 = \text{TWD } 607.16$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

Q4 2029: OP window Q4 2029 to Q3 2030: $(115,852.09 + 122,330.93 + 129,575.97 + 136,943.18) \times 6 / 4,724.215 = \text{TWD } 641.00$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

Q1 2030: OP window Q1 2030 to Q4 2030: $(122,330.93 + 129,575.97 + 136,943.18 + 144,433.12) \times 6 / 4,724.215 = \text{TWD } 677.30$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A, credit cost and minority OP claim.

The final three operating quarters support earlier four-quarter windows and have no standalone ceiling. Arithmetic uses unrounded inputs; displayed components are rounded for readability.

The August presentation states the parent ECB was fully converted by the August 3 announcement, with shares expected to rise from 4,500M to approximately 4,720M. Use $4,720 + \text{Q2 employee-compensation dilution } 4.215 = 4,724.215\text{M}$ forward diluted shares. The 4,720M figure is rounded management disclosure, not a verified final registry count. No buyback or future equity issue is assumed. Separately, Nuvoton ownership was 48.97% after its own bond conversion; this is an earnings-ownership issue, not another 51.03% dilution of Winbond shares. [W1,W2]

Mechanical share sensitivity

Q1 2028 denominator	Ceiling/share
Current model	471.88
5% more shares	449.41
10% more shares	428.98

6. Cross-book relationships and monitoring

Winbond <-> Nanya: specialty DRAM contract prices and supply additions; Winbond also has NOR/SLC and logic exposure.

Winbond <-> SNDK/Kioxia/MU/Samsung/SK hynix: broad NAND scarcity is supportive, but SLC/code storage has different customers, density and ASP.

Winbond <-> AUO/industrial and edge-AI suppliers: qualification and device content widen demand without implying direct disclosed customer relationships.

Information processing: primary segment/monthly data affects the operating path; conversion affects denominator; minority ownership affects attributable OP. This deep dive changes no other company Base.

7. Scored risks and re-rate tests

Risk class	Score	Failure channel
Competition	-1	Chinese NOR/DRAM supply and qualified incumbents can lower ASPs or take sockets.
Execution	-1	16nm/24nm yields, CUBE design-ins and Module B commissioning may lag; do not pull B into 2027.
Demand / product concentration	-1	Memory contributes most profit; elevated pricing can trigger customer redesign or destocking.
Capital allocation / dilution	-1	Fab spending, parent ECB shares and Nuvoton minority dilution constrain per-share value.

Dated triggers and counters

Checkpoint	Positive evidence	Counter / model action
1. September/October 2026	September revenue near modeled 27,855; Q3 segment mix and costs disclosed	Replace residual CMS ASP calibration with actual segment shipment/price mix.
2. 2027 16nm and 24nm ramp	Qualified bits and NAND annual growth confirm management direction	Weak yield or demand reduces volume and raises unit cost together.
3. 2027-28 CUBE/Si-Cap	Named qualification, design-in and paid production progress	60% weight rises on orders, falls or shifts on delays; do not create duplicate revenue.
4. 2029 Module B	Tool installation and late-year qualified output	Move the 80%-weighted increment if schedule shifts; no 2027 B revenue.
5. Every reporting period	Current diluted register and Nuvoton ownership reconciled	Recalculate shares/minority claim and do not value 100% of subsidiary profit.

NOR/SLC mix, separate Flash gross margins, customer wallet shares and full management Q&A are unavailable. Ownership OP is a transparent non-GAAP proxy: Nuvoton standalone/segment eliminations and its own minority holdings prevent exact attribution from these disclosures. Logic OP proxy = consolidated logic revenue x (logic GM - allocated R&D - allocated SG&A); subtract 51.03%. R&D/SG&A are already inside consolidated expenses and are not deducted twice. Other/NCI in model tables combines a credit-loss allowance and this minority adjustment, which can be negative in a loss quarter. The exact bridge is shown below. Future forecast confidence deteriorates materially beyond 12-16 months; no verified Yahoo HIGH-estimate floor is asserted.

8. Sources and snapshot

- **W1** - August 6, 2026. Winbond Q2 reviewed consolidated statements. Reviewed totals, half-year segments, Nuvoton ownership and dilution.
- **W1a** - May 2026. Winbond Q1 reviewed consolidated statements. Q1 segment revenue subtracted from H1 to derive exact Q2 segments.
- **W2** - August 6, 2026. Winbond Q2 management presentation. Memory/logic margins, shipment/pricing, capex, bond conversion, new technologies and NAND bit-growth outlook.
- **W3** - May 5, 2026. Winbond Q1 management presentation. Prior-quarter technology, shortage and market-share context.
- **W4** - September 2026. Winbond July and August monthly revenue. Unaudited reported monthly revenue; September remains estimated.
- **W5** - 2026 publication. Winbond 2025 Annual Report. Market competition and product use sections, printed pages 78-80; legacy market figures are historical.
- **W6** - August 6, 2026. CNA management interview: Kaohsiung expansion timing. Local reported management comments distinguish Module A from late-2029 Module B production.
- **W7** - September 21, 2026. Winbond market quote. TWD174 at September 21 14:30; header only.
- **W8** - Accessed September 21, 2026. Winbond product innovation disclosure. CUBE custom-memory and flash engineering scope, not accelerator-HBM market-share evidence.
- **U1** - Provided in this conversation. User-provided TrendForce NAND and BofA memory outlook. NAND price +23% Q3 2026, +20% Q4 and +15% Q1 2027. Broad NAND forecasts are only a proxy for SLC pricing. 2028 NAND -13% and DRAM -5% are from the supplied BofA image.
- **U2** - September 21, 2026. User valuation instruction. Use 6x operating profit for Nanya and Winbond; existing Base scores unchanged.

Model snapshot: September 21, 2026. 2344.TW. Framework 6.0.2. Base 48/60 (80%). Eighteen modeled operating quarters and fifteen complete forward-four-quarter ceilings. Current full PDF and synchronized Markdown: G:/My Drive/1amInvesting/Artifacts/2344TW_WinbondElectronics_DeepDive.pdf and .md. The companion company worksheet uses this same model and links its ceiling series to the master portfolio.